ACT / ALOHA · arXiv 2304.13705 · RSS 2023

Empiricist

11+3 · weight 3

k-sweep · TE · CVAE

Transfer Cube first

Oct 12 paper club · VLA Architectures

Spine: action chunking as motor contract — TE ≠ receding ≠ open-loop ≠ soft-inpaint

Zhao, Kumar, Levine, Finn · Stanford / Berkeley / Meta

Empiricist hypotheses · weight 3

H-chunk	Mid chunk (~2 s, k≈50–100) beats k=1 and episode-long
H-ensemble	TE = control-time filter; +~3 pp / lower jerk @ 50 Hz
H-CVAE	z matters iff demos multimodal (human); ≈noop on scripted
H-alphabet	Absolute joint targets ≫ deltas for this PID stack
H-lineage	commit_seconds = executed/Hz is the transferable knob

Falsifiers: k=1≈k=100 · TE always hurts · CVAE helps scripted equally · deltas ≥ targets

Effect-size tattoos · memorize these

Chunking

k=1 → k=100

1% → 44%

sim aggregate, TE off

CVAE (human)

with → without

35.3% → 2%

scripted: ≈ no change

Temporal ensemble

ACT gain

+3.3 pp

BC-ConvMLP +4% · VINN hurt

Teleop Hz

50 Hz vs 5 Hz

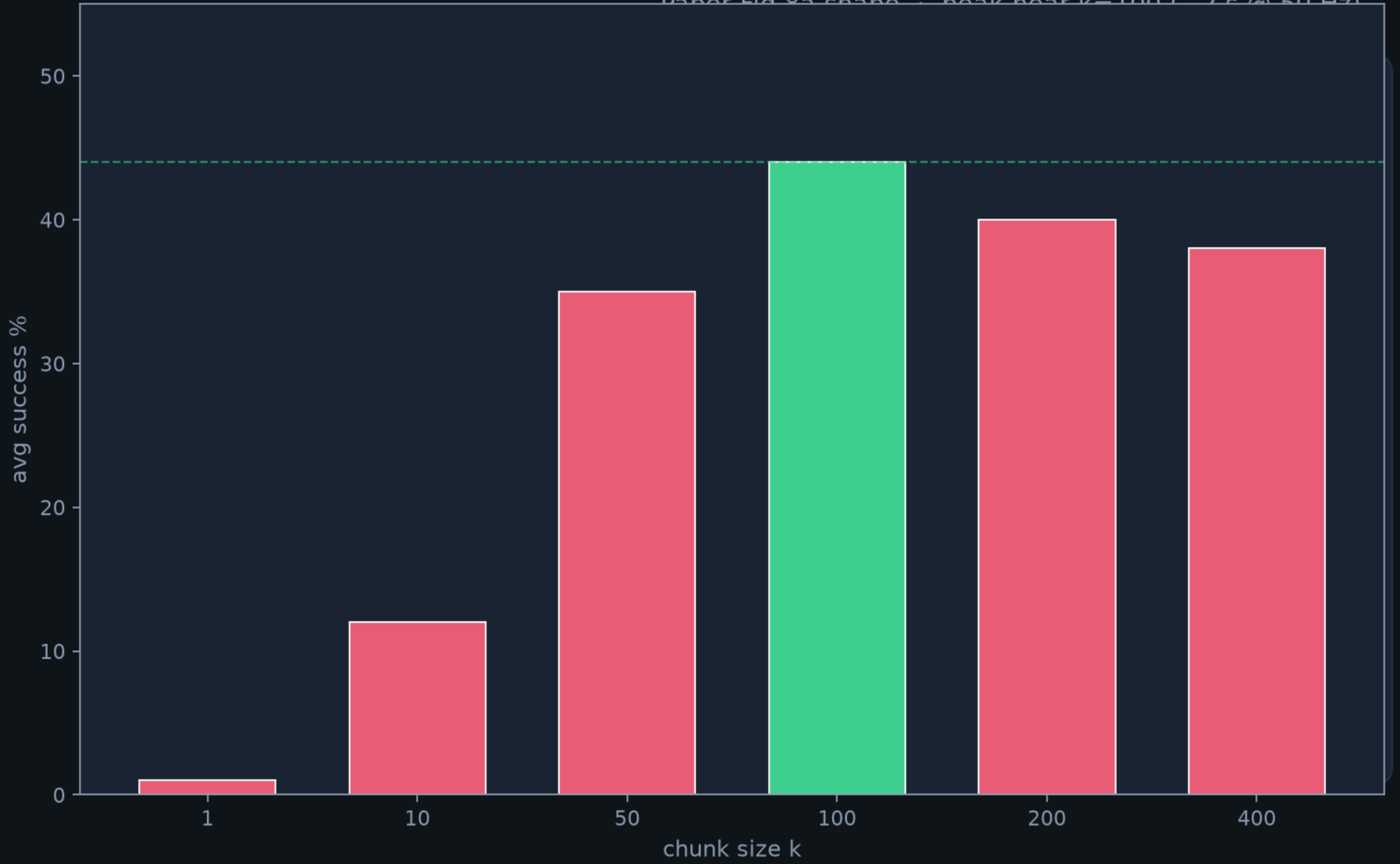
~62% slower

user study $p<0.001$

k-sweep (no TE) · sim aggregate

k=1 → 100 lifts ~1% → 44%

Paper Fig 8a shape · peak near k=100 (~2 s @ 50 Hz)



P0 grid: k ∈ {1,10,50,100,200,400} · TE off for pure chunk curve

k=1

~1% — single-step BC
compounding death

k=100 ★

~44% sweet spot
~2 s @ 50 Hz

k=200–400

slight dip
near open-loop hard

Experiment priority matrix

P0 Y	k ∈ {1,10,50,100,200,400} (TE off)	Peak ~100; k=1 collapses
P0 Y	TE on/off at best k	+~3 pp / lower jerk
P0 Partial	CVAE on/off · human vs scripted	Critical on human only
P1	Chunking on BC-MLP; target vs delta; L1 vs L2	Generalize chunk claim
Defer	Full real ALOHA 6-task / paper seeds	Needs hardware

Pass bar: qualitative Fig-8 shape on sim — not Table I real-robot parity

Action chunking vs Temporal Ensembling

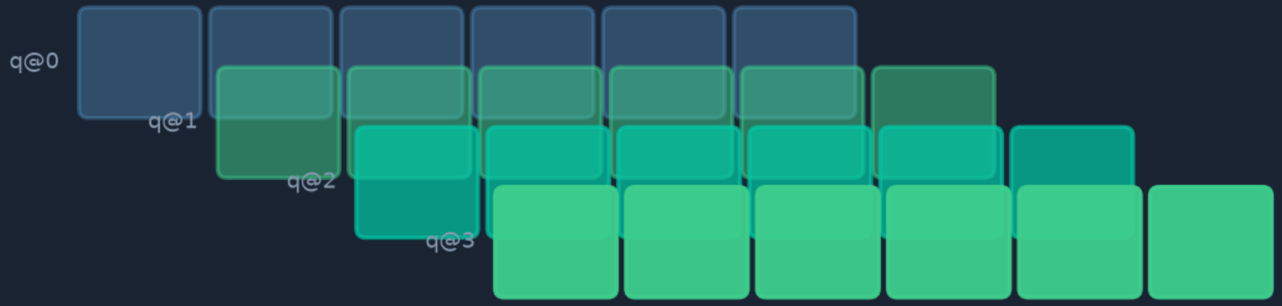
Naïve open-loop every k steps vs every-step query + exp-weighted overlap

Naïve chunking (open-loop)

observe → predict k → execute all k → replan · jerky at boundaries

ACT Temporal Ensembling

Query every tick · buffer overlapping preds for same absolute t · $a_t = \Sigma w_i \hat{A}_t[i] / \Sigma w_i$ · $w_i = \exp(-m \cdot i)$



Same-t fusion

$w = [e^{0}, e^{-m}, e^{-2m}, \ldots]$

recent obs → faster (small m)

+~3.3 pp ACT (paper)

Control schemes — do NOT conflate

Chunking is shared · TE ≠ receding ≠ open-loop ≠ soft-inpaint

Scheme	Mechanism	Commit / loop
ACT TE	Every-step query; exp-avg overlapping preds for same t	ensembled 1-step @ 50 Hz
DP receding	Predict Tp, exec Ta < Tp, replan	Ta≈8 @ ~10 Hz
π_0 open-loop	H=50; exec prefix; NO TE (hurt early for them)	open-loop prefix @ 20–50 Hz
RTC soft-inpaint	Async; freeze prefix; soft-mask remainder	latency-aware async
BEHAVIOR’25	Predict 30 / exec 26 / keep 4 + correlated FM	receding ~30 Hz

Map: k/50 ↔ DP Ta·Δt ↔ π_0/Comet H/Hz · Defaults: k=100 (~2s @ 50 Hz); L1+βKL; test z=0

Horizon map · normalize by commit seconds

ACT	k=100 @ 50 Hz	**~2.0 s chunk**	TE → 1-step ensembled
DP	Tp=16, Ta=8 @ 10 Hz	**~0.8 s commit**	receding replan
π_0	H=50 @ 20–50 Hz	**~1.0–2.5 s**	open-loop prefix
BEHAVIOR’25	30 / exec 26 @ ~30 Hz	**~0.87 s**	keep 4 + inpaint

k/50 ↔ Ta·Δt ↔ H/Hz — Empiricist + Archaeologist shared language

GPU budget · quiet P0–P1 honesty

Paper ref: ~5 h / 11 GB 2080 Ti · target ≲40–60 GPU-h scout

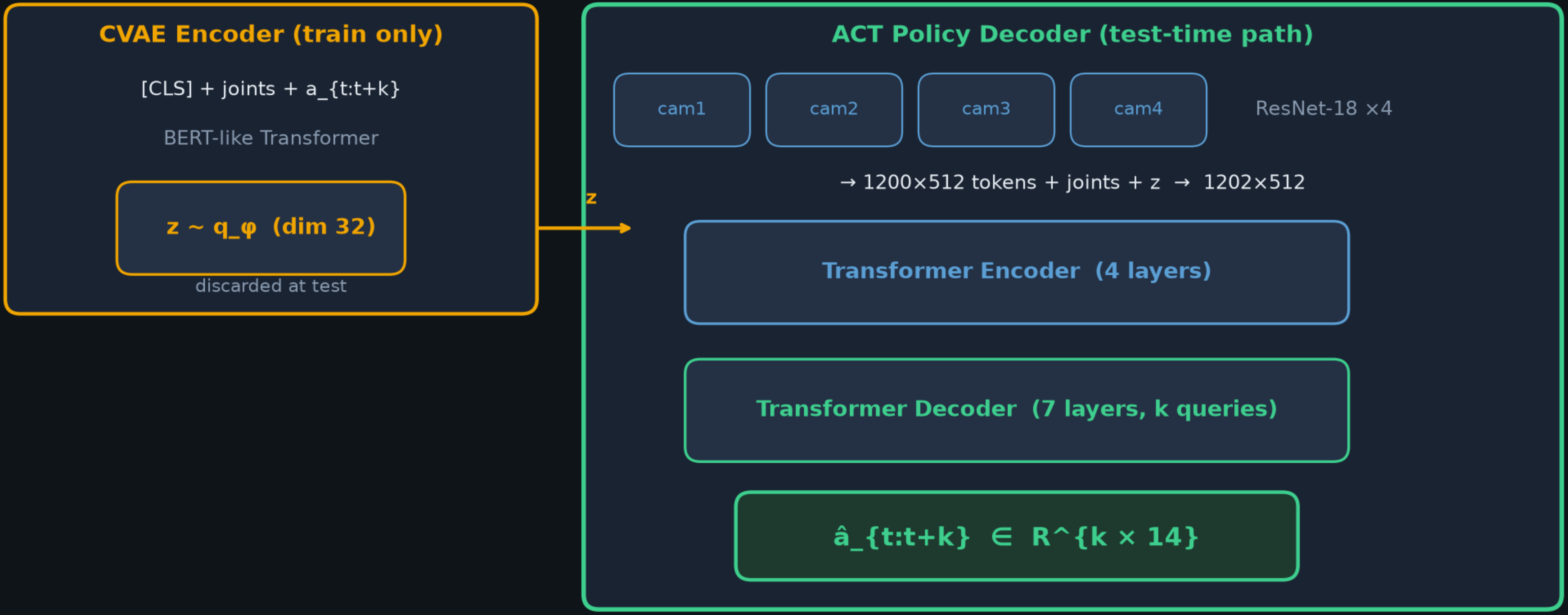
Workload	VRAM	Wall	Smoke
Transfer Cube default k=100	8–12 GB	4–8 h A100	Y
k-sweep 6 values, 1 seed	8–12 GB	24–48 h array	Y
TE × CVAE 2×2 @ best k	8–12 GB	16–32 h	Y
Human-demo CVAE ablate	8–12 GB	+1 day	Partial
Full real 6-task ALOHA	robot	days+teleop	N — defer

Minimal matrix: {k: 1|10|100|400} × {TE: on|off} × {CVAE: on|off}

Transfer Cube scripted first · log device + max_memory_allocated + (k,TE,CVAE,Hz,ms)

ACT architecture · CVAE + Transformer

~80M · L1 + β KL (β=10) · k×14 absolute joints · z=0 at test



Obs: 4× RGB 480×640 + 14 joints · Action: absolute joint targets (not delta) · PID @ Dynamixel

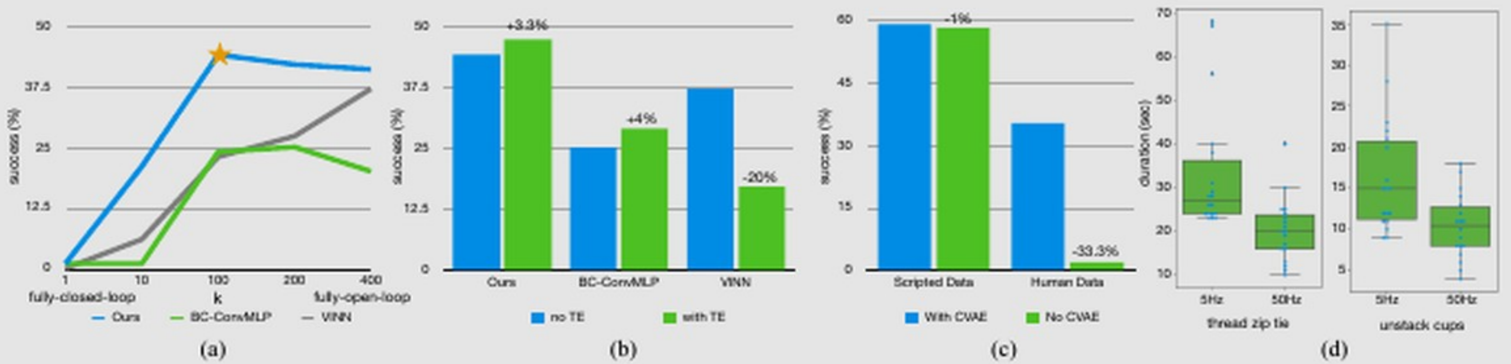


Fig. 8: *(a)* We augment two baselines with action chunking, with different values of chunk size k on the x-axis, and success rate on the y-axis. Both methods significantly benefit from action chunking, suggesting that it is a generally useful technique. *(b)* Temporal Ensemble (TE) improves our method and *BC-ConvMLP*, while hurting *VINN*. *(c)* We compare with and without the CVAE training, showing that it is crucial when learning from human data. *(d)* We plot the distribution of task completion time in our user study, where we task participants to perform two tasks, at 5Hz or 50Hz teleoperation frequency. Lowering the frequency results in a 62% slowdown in completion time.

(c), we visualize the success rate aggregated across 2 simulated tasks, and separately plot training with scripted data and with human data. We can see that when training on scripted data, the removal of CVAE objective makes almost no difference in performance, because dataset is fully deterministic. While for human data, there is a significant drop from 35.3% to 2%. This illustrates that the CVAE objective is crucial when learning from human demonstrations.

C. Is High-Frequency Necessary?

Lastly, we conduct a user study to illustrate the necessity of high-frequency teleoperation for fine manipulation. With the same hardware setup, we lower the frequency from 50Hz to 5Hz, a control frequency that is similar to recent works that use high-capacity deep networks for imitation learning [7, 70]. We pick two fine-grained tasks: threading a zip cable tie and un-stacking two plastic cups. Both require millimeter-level precision and closed-loop visual feedback. We perform

learning algorithm *ACT*. The synergy between these two parts allows us to learn fine manipulation skills directly in the real-world, such as opening a translucent condiment cup and slotting a battery with a 80-90% success rate and around 10 min of demonstrations. While the system is quite capable, there exist tasks that are beyond the capability of either the robots or the learning algorithm, such as buttoning up a dress shirt. We include a more detailed discussion about limitations in Appendix F. Overall, we hope that this low-cost open-source system represents an important step and accessible resource towards advancing fine-grained robotic manipulation.

Acknowledgement

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Left: Cube Transfer. Transfer the red cube to the other arm. The right arm touches *(#1)* and grasps *(#2)* the red cube, then hands it to the left arm.
Right: Bimanual Insertion. Insert the red peg into the blue socket. Both arms grasp *(#1)*, let socket and peg make contact *(#2)* and insertion.

Fig. 7: *Simulated Task Definitions.* For each of the 2 simulated tasks, we illustrate the initializations and the subtasks.

	Cube Transfer (sim)			**Bimanual Insertion** (sim)			**Slide Ziploc** (real)			**Slot Battery** (real)		
	Touched	Lifted	Transfer	Grasp	Contact	Insert	Grasp	Pinch	Open	Grasp	Place	Insert
BC-ConvMLP	34 | 3	17 | 1	1 | 0	5 | 0	1 | 0	1 | 0	0	0	0	0	0	0
BeT	60 | 16	51 | 13	27 | 1	21 | 0	4 | 0	3 | 0	8	0	0	4	0	0
RT-1	44 | 4	33 | 2	2 | 0	2 | 0	0 | 0	1 | 0	4	0	0	4	0	0
VINN	13 | 17	9 | 11	3 | 0	6 | 0	1 | 0	1 | 0	28	0	0	20	0	0
ACT (Ours)	**97 | 82**	**90 | 60**	**86 | 50**	**93 | 76**	**90 | 66**	**32 | 20**	**92**	**96**	**88**	**100**	**100**	**96**

TABLE I: Success rate (%) for 2 simulated and 2 real-world tasks, comparing our method with 4 baselines. For the two simulated tasks, we report [training with scripted data | training with human data], with 3 seeds and 50 policy evaluations each. For the real-world tasks, we report training with human data, with 1 seed and 25 evaluations. Overall, ACT significantly outperforms previous methods.

	Open Cup (real)			**Thread Velcro** (real)			**Prep Tape** (real)				**Put On Shoe** (real)			
	Tip Over	Grasp	Open Lid	Lift	Grasp	Insert	Grasp	Cut	Handover	Hang	Lift	Insert	Support	Secure
BeT	12	0	0	24	0	0	8	0	0	0	12	0	0	0
ACT (Ours)	**100**	**96**	**84**	**92**	**40**	**20**	**96**	**92**	**72**	**64**	**100**	**92**	**92**	**92**

TABLE II: Success rate (%) for the remaining 3 real-world tasks. We only compare with the best performing baseline BeT.

tape dispenser. Especially from the top view, it is challenging to localize the velcro tie because of the small projected area.

the handover, and the left gripper should always move to a position that can grasp the tape, instead of trying to memorize where exactly the handover happens, which can vary across

Scout k · TE · CVAE on Transfer Cube

Tattoo: k=1→100 = 1%→44% · CVAE human 35.3%→2%
Budget 8–12 GB · ≲40–60 GPU-h · pass = Fig-8 shape, not Table-I parity

Q&A · keep the spine clear

Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

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[1] Stanford University [2] UC Berkeley [3] Meta

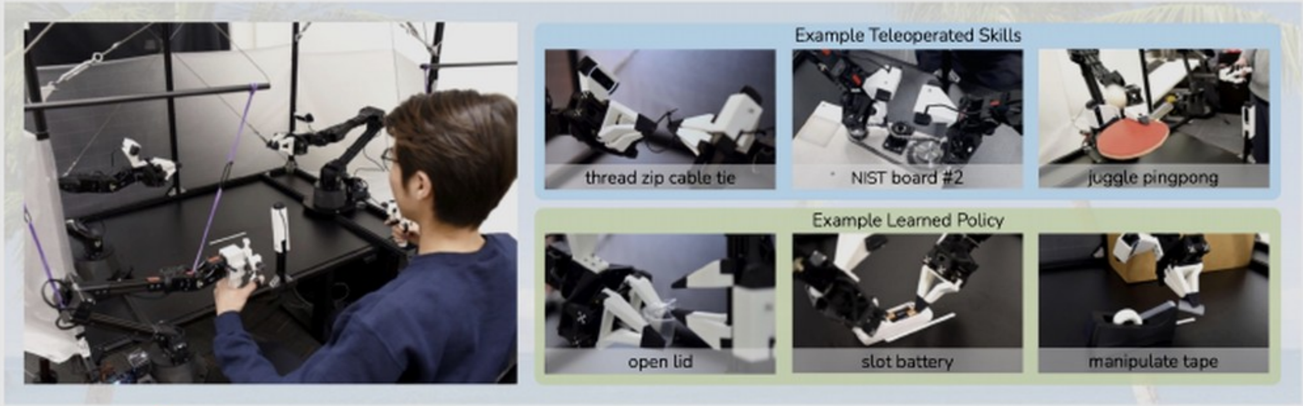

Fig. 1: *ALOHA : A Low-cost Open-source Hardware System for Bimanual Teleoperation*. The whole system costs <$20k with off-the-shelf robots and 3D printed components. *Left:* The user teleoperates by backdriving the leader robots, with the follower robots mirroring the motion. *Right: ALOHA* is capable of precise, contact-rich, and dynamic tasks. We show examples of both teleoperated and learned skills.

***Abstract*—Fine manipulation tasks, such as threading cable ties or slotting a battery, are notoriously difficult for robots because they require precision, careful coordination of contact forces, and closed-loop visual feedback. Performing these tasks typically requires high-end robots, accurate sensors, or careful calibration, which can be expensive and difficult to set up. *Can learning enable low-cost and imprecise hardware to perform these fine manipulation tasks?* We present a low-cost system that performs end-to-end imitation learning directly from real demonstrations, collected with a custom teleoperation interface. Imitation learning, however, presents its own challenges, particularly in high-precision domains: errors in the policy can compound over time, and human demonstrations can be non-stationary. To address these challenges, we develop a simple yet novel algorithm, *Action Chunking with Transformers (ACT)*, which learns a generative model over action sequences. ACT allows the robot to learn 6 difficult tasks in the real world, such as opening a translucent condiment cup and slotting a battery with 80-90% success, with only 10 minutes worth of demonstrations. Project website: tonyzhaozh.github.io/aloha**

off the table. Next, one of the right fingers approaches the cup from below and pries the lid open. Each of these steps requires high precision, delicate hand-eye coordination, and rich contact. Millimeters of error would lead to task failure.

Existing systems for fine manipulation use expensive robots and high-end sensors for precise state estimation [29, 60, 32, 41]. In this work, we seek to develop a low-cost system for fine manipulation that is, in contrast, accessible and reproducible. However, low-cost hardware is inevitably less precise than high-end platforms, making the sensing and planning challenge more pronounced. One promising direction to resolve this is to incorporate learning into the system. Humans also do not have industrial-grade proprioception [71], and yet we are able to perform delicate tasks by learning from closed-loop visual feedback and actively compensating for errors. In our system, we therefore train an end-to-end policy that directly maps RGB images from commodity web cameras to the actions.